24-311: Numerical Methods

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It's Math, not Rocket Science

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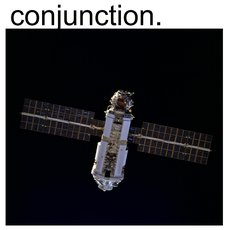
Background and Motivation

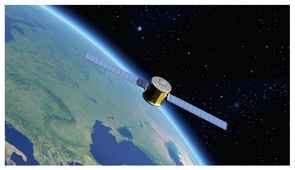
What is the broader context?

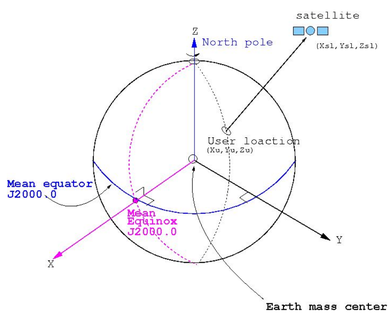
- Rocket trajectories are incredibly important to determining how, where, and when to launch rockets and satellites. Billions of people depend on the information sent through this technology which makes these calculations crucial.

How does your work fit in?

- Our work provides a model for trajectory that can be used as a low cost and rapid model.
- This work also provides an example for orbit that can be used for many different satellite calculations in conjunction.







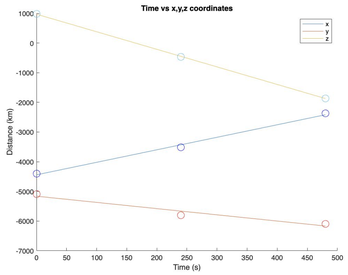
For our project we used the EME 2000 coordinate system as used by NASA and the data we based our code on. This structures the Earth as the center of our system.

Data Collection Methods

How did you collect your data?

The data we collected was from a NASA ISS publicly available dataset. We cannot cite an exact source as the data was recently removed.

The graph displays the position state vectors of the Zarya Module provided to us through the dataset., We have plotted the launch trajectory data and will use it to compare our estimation using the Runge Kutta Method



Numerical Method/Equations of Motion

What numerical methods/EoMs did you use?

- We used the Runge-Kutta Method, or RK4, a method used to solve initial value differential equations. Our equations of motion implemented thrust[1], gravity and drag into Newton's Second Law.

How do they work?

- Equation of Motion:

$$\sum F = ma = F_{drag} + F_g + F_{thrust}$$

$$F_{thrust} = \dot{m} g I$$

$$F_{drag} = \tfrac{1}{2} \rho C_d A_c v^2$$

$$F_g = G m_{earth} / r^2$$

h = Time Step

Runge–Kutta method

(1) $y' = F(x,y), \quad y_0 = f(x_0) \rightarrow y = f(x)$

(2) $y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) + O(h^5)$

$$k_1 = hF(x_n, y_n)$$

$$k_2 = hF(x_n + \frac{h}{2}, y_n + \frac{k_1}{2})$$

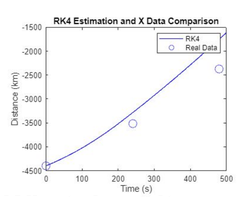
$$k_3 = hF(x_n + \frac{h}{2}, y_n + \frac{k_2}{2})$$

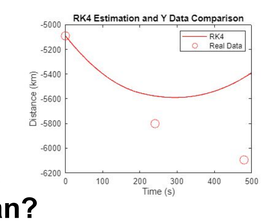
$$k_4 = hF(x_n + h, y_n + k_3)$$

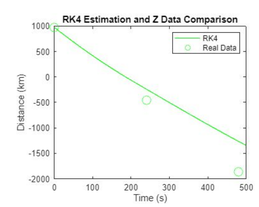
Results and Discussion

What did you find out?

- Using the RK4 method, we found the trajectory of the Zarya module over the course of 480 seconds, the predicted amount of time that it would take for the Proton-K rocket to reach orbit. Using this information, we then compared our model to the original data set and saw that the error was significant.

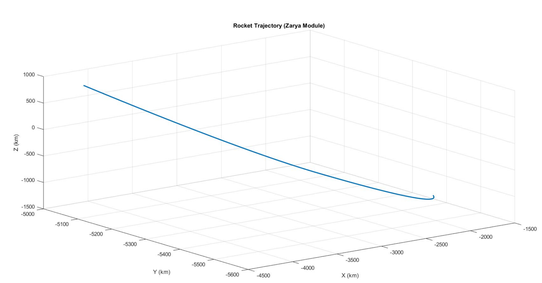


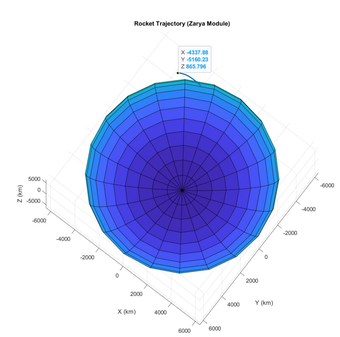




What does it mean?

- As we can observe from our error graphs, the predicted trajectory of the rocket using RK4 was similar to the actual trajectory of the rocket. This difference can be attributed to the simplification of our problem, which we believe contributed to our error. Certain details including the continuous trajectory adjustment from boosters as well as the changes entering into orbit and multi-stage ignition cycles were left out for the scope of this project.





Conclusions

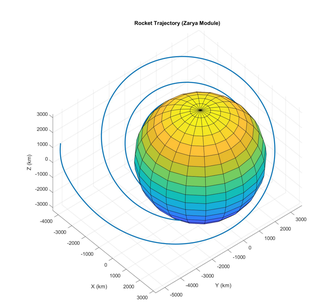
Our results here speak to both the robustness, and the limitations of the Runge-Kutta methods in trajectory planning.

Even with many assumptions and simplifications, the methods still allowed us to generate a trajectory for the Proton-K rocket with relatively low error, using only the model and initial conditions.

However, the importance of a good model is also to be noted. Simplified versions of the thrust vector provided by the rocket, and no data with regard to the rocket's vector's direction introduced unavoidable error into the model, exaggerated by the RK4 integration technique.

Future Work

Models like this one can be used to predict the trajectory of spacecraft without access to thrust vector data. This could be useful as a fallback method for future aerospace missions if communication with the craft is lost.



References

1) "Rocket Thrust Equations." NASA, NASA, www.grc.nasa.gov/www/k-12/airplane/rktthsum.html. Accessed 21 Apr. 2025.5t`3
2) "Runge-Kutta Method." From Wolfram MathWorld, mathworld.wolfram.com/Runge-Kutta Method.html. Accessed 21 Apr. 2025.